

SURV613/SURVMETH640: Machine Learning for Social Science

Tues 3:45pm – 6:15pm
Winter/Spring 2025

Instructor: Brian Kim

Email: kimbrian@umd.edu

Office Hours: By appointment. [Schedule here](#).

Course Description: This course provides an introduction to supervised statistical learning techniques such as decision trees, random forests and boosting and discusses their potential application in the social sciences. These methods focus on predicting an outcome Y based on some learned function $f(X)$ and therefore facilitate new research perspectives in comparison with traditional regression models, which primarily focus on causation. Predictive methods also provide a valuable extension to the empirical social scientists' toolkit as new data sources become more prominent. In addition to introducing supervised learning methods, the course will include practical sessions to exemplify how to tune and evaluate prediction models using the statistical programming language R.

Learning Outcomes: After taking this course, student will be able to:

- Implement the full machine learning workflow, including data preparation, model building, validation, and testing.
- Describe the bias-variance tradeoff and how it affects the machine learning process.
- Use R to fit a wide variety of machine learning models for both regression and classification.
- Critically evaluate models based on performance metrics as well as bias and fairness measures.

Class Meetings: This course uses a flipped classroom design. In this course, you are responsible for watching video recorded lectures and going through the readings, then attending class meetings where students have the chance to discuss the materials from a unit with the instructor. In general, the in-class lecture will be focused primarily on student questions as well as walking through R code examples.

In preparation for class meetings, students are expected to watch the lecture videos and go over the readings before the start of the meeting. In addition, **students must post questions or comments about the materials covered in the videos and readings of the week to the instructor in the forum before the meetings**. If you do not have questions, you may respond to another student, or comment about possible topics of interest to you, such as different applications of the tools learned in class. The deadline for posting questions is three hours before the online meeting. **Posting weekly will account for 10% of your final grade.**

Prerequisites: Although this course will include a short introduction to R, some prior programming knowledge in R is assumed. If necessary, students are encouraged to work through one or more R tutorials prior to or during the first weeks of the course. Some resources can be found here:

<https://www.rstudio.com/online-learning/#R>
<https://rstudio.cloud/learn/primers>
<http://www.statmethods.net/>
<https://swirlstats.com/>

Course Grades: Your grade in this class will consist of three parts: assignments, forum posts, machine learning project. A missing submission will be scored as zero. The distribution for the final grade will be:

- 50% Assignments
- 10% Forum posts
- 40% Final paper (including proposal).

Course Objectives: At the completion of this course, students will have a profound understanding of tree-based prediction methods and the machine learning perspective on statistical modeling. Students will learn the computational skills to apply and evaluate these methods using R.

Required Textbooks:

Hastie, T., Tibshirani, R., and Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction. New York, NY: Springer.

James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). An Introduction to Statistical Learning. New York, NY: Springer.

Recommended Textbooks:

A few readings will come from these textbooks, but you should be able to find these readings online as well.

Foster, I., Ghani, R., Jarmin, R. S., Kreuter, F., and Lane, J. (Eds.). (2017). Big Data and Social Science: A Practical Guide to Methods and Tools. Boca Raton, FL: CRC Press Taylor & Francis Group.

Hastie, T., Tibshirani, R., Wainwright, M. (2015). Statistical Learning with Sparsity: The Lasso and Generalizations. Boca Raton, FL: Chapman & Hall/CRC.

Kuhn, M. and Johnson, K. (2013). Applied Predictive Modeling. New York, NY: Springer.

Course Policies & Requirements: Students are required to have a computer with R (www.R-project.org) and RStudio (www.rstudio.com) installed. Students are asked to attend classes on time and remain through the entire class. Regular attendance is highly recommended as the graded exercises build heavily on these sessions. Students are required to inform the instructor about absences that may conflict with submitting graded course work in a timely manner. Further information on attendance policies can be found at the University of Maryland, Office of Faculty Affairs website. <https://faculty.umd.edu/teach/#attend>

Academic Honesty Policy: Clear definitions of the forms of academic misconduct, including cheating and plagiarism, as well as information about disciplinary sanctions for academic misconduct, may be found at the University of Maryland, Office of the President's website. <http://www.president.umd.edu/policies/docs/III-100A.pdf>

Disability Accommodation: In order to receive services you must contact the Accessibility and Disability Service (ADS) office to register in person for services. Please call the office to set up an appointment to register with an ADS counselor. Contact the ADS office: <https://www.counseling.umd.edu/ads/>.

Student Health and Medical Emergency: For mental and physical health resources, visit <http://www.health.umd.edu/>.

Course Outline (Tentative):

Textbook Abbreviations:

- *ISL: Introduction to Statistical Learning*
- *ESL: Elements of Statistical Learning*
- *APM: Applied Predictive Modeling*
- *SLS: Statistical Learning with Sparsity*
- *BDSS: Big Data and Social Science*
- *IML: Interpretable Machine Learning*

Date	Readings	Content
1/14	<i>None</i>	Introduction; R and RStudio
1/21	<i>ISL 2, 5</i>	ML Basics: Bias-Variance Trade-Off, Cross-Validation <i>Assignment 1 assigned (due 11:59 p.m. on 2/3)</i>
1/28	<i>ESL 2.3, APM 11, BDSS 7.7</i>	Performance measures; K-Nearest Neighbors
2/4	<i>ISL 6.1, 6.2</i>	Regularized Regression; Model tuning <i>Assignment 2 assigned (due 11:59 p.m. on 2/17)</i>
2/11	<i>ISL 8.1, Kopf et al. (2010)</i>	Decision Trees: CART, CTREE, Model-based recursive partitioning
2/18	<i>ISL 8.2, ESL 15</i>	Ensemble Methods: Bagging, Random Forests, Extra Trees <i>Assignment 3 assigned (due 11:59 p.m. on 3/10)</i>
2/25	<i>ESL 10.1-2, 10.9-10</i>	Boosting I: Adaboost, Gradient Boosting, XGBoost
3/4	Michigan Spring Break (No Class)	
3/11	<i>ESL 11</i>	Neural Networks: tensorflow and keras <i>Assignment 4 assigned (due 11:59 p.m. on 3/31)</i>

3/18 **Maryland Spring Break (No Class)**

3/25 [IML 8, 9](#) Interpretable ML

4/1 [BDSS 11](#) Bias and Fairness
Assignment 5 assigned (due 11:59 p.m. on 4/14)

4/8 *Breiman (1996),
Chawla et al. (2002)* ML Techniques

4/15 *ESL 14* Unsupervised Learning

4/22 *See Canvas* Generative AI

4/29 **Final Paper due (11:59pm)**